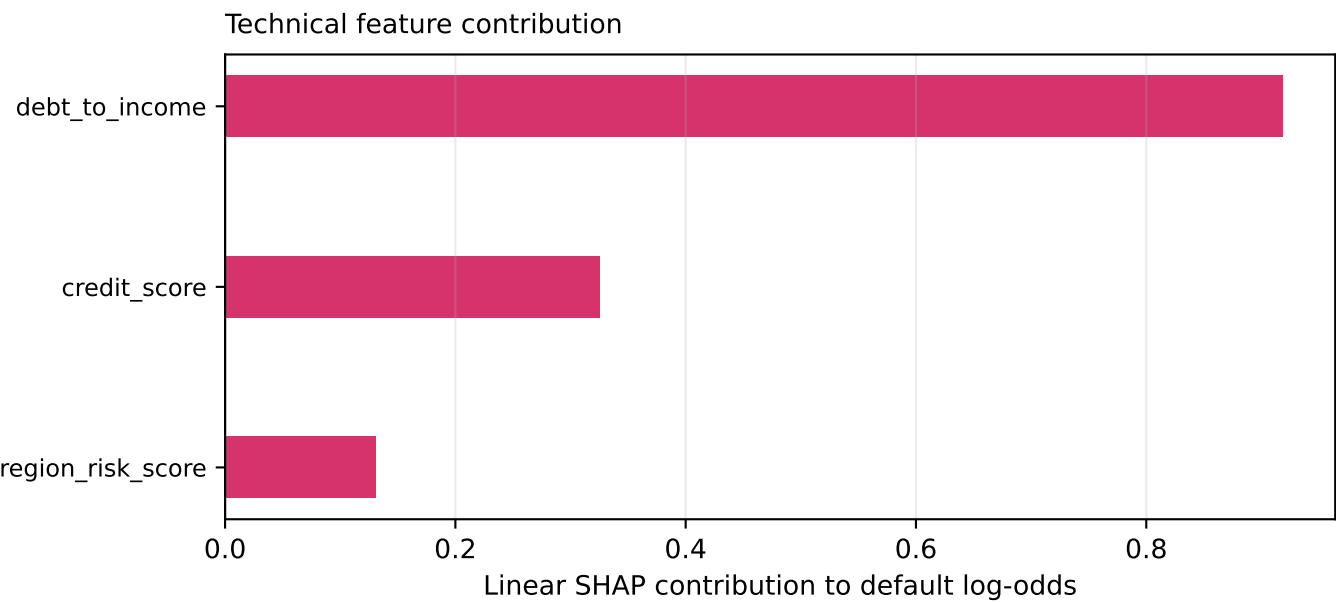


Task 3 - SHAP Explainability

APP-00011

High debt-to-income ratio / Low credit score / Elevated regional portfolio risk

Model score: 0.2834 Reject if score >= 0.20 Status: rejected



Consumer-Facing Reason	Technical Feature	SHAP Log-Odds
High debt-to-income ratio	debt_to_income	+0.9182
Low credit score	credit_score	+0.3253
Elevated regional portfolio risk	region_risk_score	+0.1306

How to read this page

Consumer-facing reason <- technical feature <- positive SHAP-style contribution.

Model link: baseline log-odds + feature contributions -> applicant log-odds -> default score.

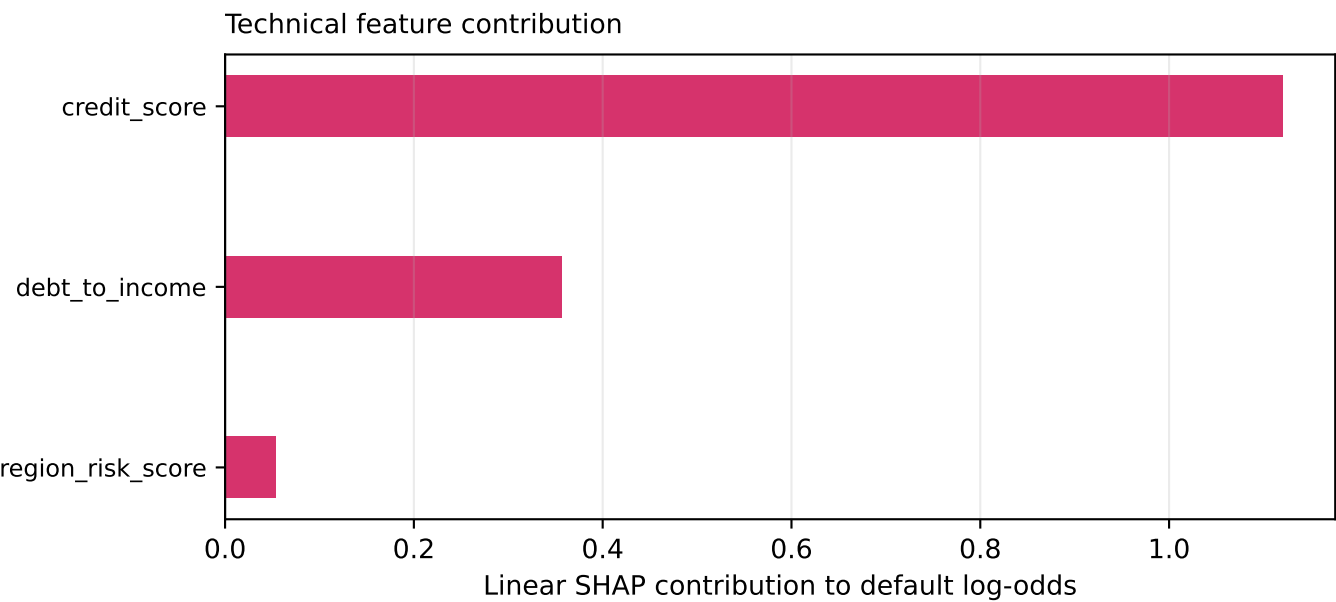
Lower-risk features are omitted from rejection explanations; these are explanation evidence, not legal advice.

Task 3 - SHAP Explainability

APP-00013

Low credit score / High debt-to-income ratio / Elevated regional portfolio risk

Model score: 0.2535 Reject if score ≥ 0.20 Status: rejected



Consumer-Facing Reason	Technical Feature	SHAP Log-Odds
Low credit score	credit_score	+1.1200
High debt-to-income ratio	debt_to_income	+0.3562
Elevated regional portfolio risk	region_risk_score	+0.0536

How to read this page

Consumer-facing reason <- technical feature <- positive SHAP-style contribution.

Model link: baseline log-odds + feature contributions -> applicant log-odds -> default score.

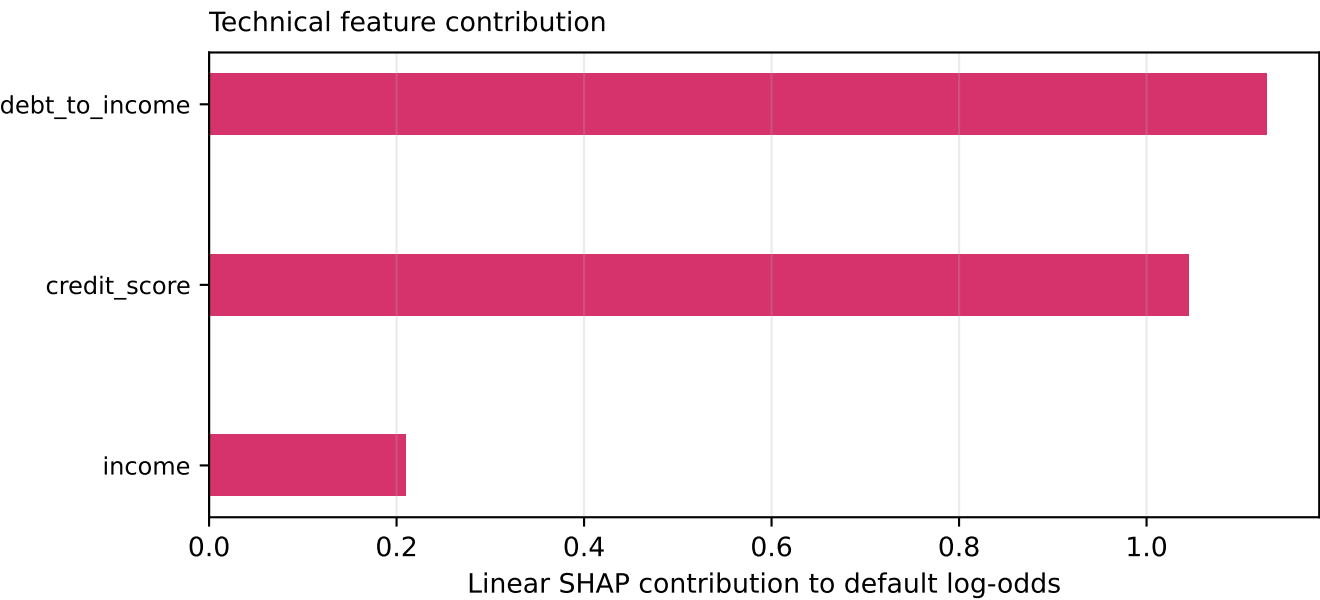
Lower-risk features are omitted from rejection explanations; these are explanation evidence, not legal advice.

Task 3 - SHAP Explainability

APP-00014

High debt-to-income ratio / Low credit score / Income insufficient for requested loan

Model score: 0.5230 Reject if score >= 0.20 Status: rejected



Consumer-Facing Reason	Technical Feature	SHAP Log-Odds
High debt-to-income ratio	debt_to_income	+1.1278
Low credit score	credit_score	+1.0452
Income insufficient for requested loan	income	+0.2095

How to read this page

Consumer-facing reason <- technical feature <- positive SHAP-style contribution.

Model link: baseline log-odds + feature contributions -> applicant log-odds -> default score.

Lower-risk features are omitted from rejection explanations; these are explanation evidence, not legal advice.

Task 3 - SHAP Methodology Note

What is being explained

The SHAP-style pages explain why selected rejected applicants received high predicted default-risk scores. They connect the model's technical feature contributions to consumer-facing adverse-action reason categories.

Why only positive contributions are shown

For an adverse-action explanation, the relevant features are those that pushed the decision toward rejection. Features that lowered default risk are useful for model debugging, but including them in a rejection-reason table would confuse the explanation given to management or consumers.

How technical explanation becomes governance evidence

The technical feature contribution is not copied directly into a consumer notice. It is mapped into a clearer reason category, such as high debt-to-income ratio or low credit score. Compliance and legal reviewers would still need to validate the final notice language.

Boundary

SHAP-style explanations improve transparency, but they do not prove legal compliance and do not replace human review. The tool treats them as evidence for governance and adverse-action reason quality, not as automatic legal conclusions.



Selected Top Contributions and Adverse-Action Mapping

Only positive SHAP-style contributions are shown, because the purpose is to explain factors that pushed the decision toward rejection.

Applicant ID	Consumer-Facing Reason	Technical Feature	SHAP Log-Odds	Direction
APP-00011	High debt-to-income ratio	debt_to_income	+0.9182	Increases risk
	Low credit score	credit_score	+0.3253	Increases risk
	Elevated regional portfolio risk	region_risk_score	+0.1306	Increases risk
APP-00013	Low credit score	credit_score	+1.1200	Increases risk
	High debt-to-income ratio	debt_to_income	+0.3562	Increases risk
	Elevated regional portfolio risk	region_risk_score	+0.0536	Increases risk
APP-00014	High debt-to-income ratio	debt_to_income	+1.1278	Increases risk
	Low credit score	credit_score	+1.0452	Increases risk
	Income insufficient for requested loan	income	+0.2095	Increases risk

Mapping logic: technical feature contributions support explanation, while the consumer-facing reason is the clearer adverse-action category. Lower-risk features are intentionally excluded from this table to avoid confusing the rejection explanation.